Champagne Tower

1. Aim

To write a C program to determine the amount of champagne present in a specified glass of a champagne tower using Dynamic Programming.

2. Theory

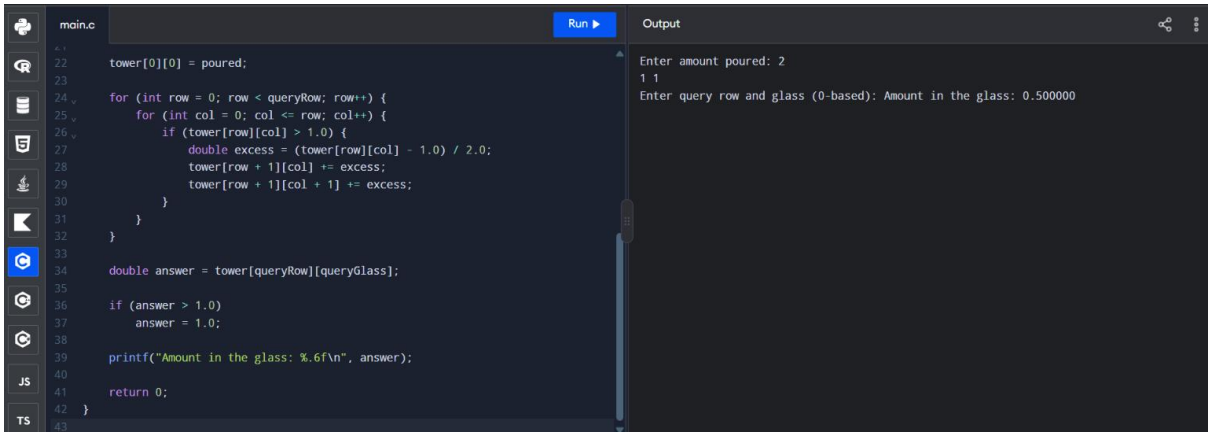
The Champagne Tower problem consists of glasses arranged in a triangular structure. The first glass is filled with champagne. Whenever a glass receives more than one unit of champagne, it retains one unit and distributes the excess equally between the two glasses directly below it.

Dynamic Programming is used to calculate how much champagne reaches each glass. The amount flowing into a glass depends on the excess champagne from the glass or glasses directly above it.

The process continues row by row until the required glass is reached. Since a glass can hold at most one unit, the final amount in the requested glass is limited to 1.

3. Algorithm

1. Start.
2. Read the poured amount, query row, and query glass.
3. Initialize the first glass with the poured amount.
4. Traverse the tower row by row.
5. If a glass contains more than one unit, calculate its excess.
6. Distribute half the excess to each of the two glasses below.
7. Continue until the required row is reached.
8. Display the amount in the specified glass, limited to 1.
9. stop



7. Complexity Analysis

Let \(r\) be the queried row.

- Time complexity: \(O(r^2)\), because the algorithm processes each glass in all rows up to row \(r\).
- Space complexity: \(O(r^2)\) for storing the tower.

8. Result

The amount of champagne present in the specified glass was calculated successfully using Dynamic Programming.